 77777

Instructor(s): *Mueller/Korytov*

PHYSICS DEPARTMENT

PHY 2049 Exam 3 April 29, 2019

Name (print): ______________________ Signature: ______________________

*On my honor, I have neither given nor received unauthorized aid on this examination.*

**YOUR TEST NUMBER IS THE 5-DIGIT NUMBER AT THE TOP OF EACH PAGE.**

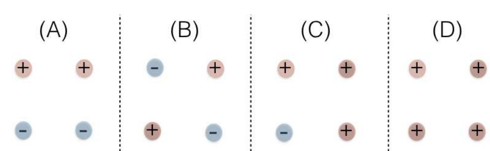
**DIRECTIONS**

(1) **<u>Code your test number</u> on your answer sheet (use 76–80 for the 5-digit number)**. Code your name on your answer sheet. **Darken circles completely (errors can occur if too light)**. Code your student number on your answer sheet.
(2) Print your name on this sheet and sign it also.
(3) Do all scratch work anywhere <u>on this exam</u> that you like. At the end of the test, this exam printout is to be turned in. No credit will be given without both answer sheet and printout with scratch work.
(4) Work the questions in any order. Incorrect answers are not taken into account in any way; you may guess at answers you don't know.
(5) If you think that none of the answers is correct, please choose the answer given that is closest to your answer.
(6) **Blacken the circle of your intended answer completely, using a number 2 pencil**. Do not make any stray marks or the answer sheet may not read properly. Completely erase all incorrect answers, or take a new answer sheet.
(7) As an aid to the examiner (and yourself), in case of poorly marked answer sheets, please circle your selected answer on the examination sheet. Please remember, however, that in the case of a disagreement, the answers on the bubble sheet count, <u>NOT</u> what you circle here. Good luck!!!

>>>>>>>>**WHEN YOU FINISH** <<<<<<<<
Hand in the answer sheet separately.

| **Constants** | | | |
|---|---|---|---|
| $e = 1.6 \times 10^{-19}$ C | $m_p = 1.67 \times 10^{-27}$ kg | $m_e = 9.1 \times 10^{-31}$ kg | $\epsilon_o = 8.85 \times 10^{-12}$ C$^2$/N·m$^2$ |
| milli (m) $= 10^{-3}$ | micro ($\mu$) $= 10^{-6}$ | nano (n) $= 10^{-9}$ | pico (p) $= 10^{-12}$ |
| $k = 1/(4\pi\epsilon_o) = 9 \times 10^9$ N·m$^2$/C$^2$ | | Sphere: $A = 4\pi R^2$ $V = \frac{4\pi}{3}R^3$ | |
| $\mu_o = 4\pi \times 10^{-7}$ T $\cdot m/A$ | | 1 eV $= 1.6 \times 10^{-19}$ J | |
| $c = 3 \times 10^8$ m/s | | | |

1. The figure shows four different quadratic configurations of four point charges $q_1$ to $q_4$. These charges are either $+1\,\mu$C or $-1\,\mu$C as indicated by the + and − signs in the drawings. Which configuration creates the largest eletrical field in the center?

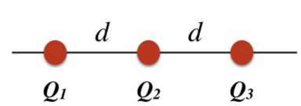



(1) only A (2) A and B (3) only D (4) only C (5) only B

2. Two identical conducting spheres, A and B, are separated by a distance $d$ which is much larger than their radius. Sphere A carries an initial charge of Q and sphere B of 2Q. A third conducting sphere, C, which is identical to spheres A and B but uncharged, is brought in contact with B and then with A, and is then taken far away. During the entire procedure, sphere C is handled with an insulating pair of tweezers. What is the electrostatic force between spheres A and B after the procedure?

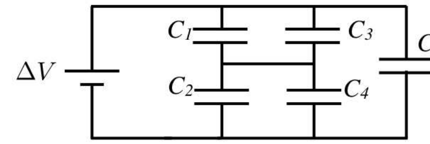
(1) $k\dfrac{Q^2}{d^2}$ (2) $k\dfrac{1.25 \cdot Q^2}{d^2}$ (3) $k\dfrac{0.625 \cdot Q^2}{d^2}$ (4) $k\dfrac{1.575 \cdot Q^2}{d^2}$ (5) $k\dfrac{2.225 \cdot Q^2}{d^2}$

3. Three charges are placed in one line with a distance of $d = 1\,$cm between $Q_1$ and $Q_2$ as well as between $Q_2$ and $Q_3$ as shown in the figure. The charges are $Q_1 = 2\,\mu$C, $Q_2 = 4\,\mu$C, and $Q_3 = -2\,\mu$C. How much work is required to swap charges $Q_1$ and $Q_2$?




(1) 1.8 J (2) −3.6 J (3) −1.8 J (4) 3.6 J (5) 5.4 J

4. Five identical capacitors of 2 $\mu$F are connected to a battery as shown in the figure. If $\Delta V = 3\,\text{V}$, what is the charge on capacitor $C_1$?

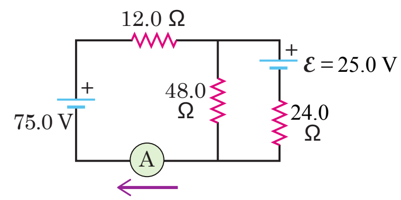



(1) 3 $\mu$C (2) 12 $\mu$C (3) 2.4 $\mu$C (4) 6 $\mu$C (5) 4.5 $\mu$C

5. A power supply line in Siberia is 2000 km long and carries a current of 4650 A. It is made from a thick conducting wire with a diameter of 5 cm and of resistivity $3.14 \cdot 10^{-8}$ $\Omega$m. How much power is lost into heat along the wire?

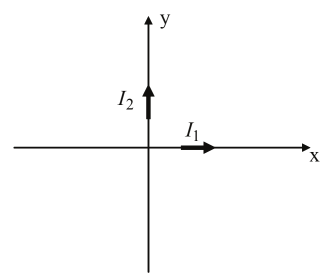
(1) 690 MW (2) 36 MW (3) 1.4 MW (4) 810 kW (5) 360 kW

6. Find the current through the ammeter in the circuit shown in the figure.



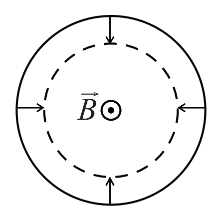

(1) 2.1 A (2) 1.6 A (3) 1.3 A (4) 0.9 A (5) 1.8 A

7. Two wires are aligned with the x- and y-axes and carry currents $I_1 = 1\text{A}$ along the x-axis and $I_2 = 2\text{A}$ along the y-axis as shown. Find the magnitude of magnetic field at point with coordinates (−1m, 2m).




(1) $5 \times 10^{-7}$ T
(2) $10^{-6}$ T
(3) $4 \times 10^{-7}$ T
(4) $8 \times 10^{-7}$ T
(5) $2 \times 10^{-6}$ T

8. A circular conducting loop is placed in a uniform 2.0-tesla magnetic field pointing out of the page, as shown. If the loop suddenly contracts from its initial radius of 0.6 m with a radial velocity of $dr/dt = -0.4$ m/s, what is the magnitude and direction of the induced current immediately after the loop starts contracting? Assume that the resistance of the loop, 0.2 $\Omega$, is not affected by the contraction.




(1) 15 A counterclockwise
(2) 15 A clockwise
(3) 7.5 A counterclockwise
(4) 7.5 A clockwise
(5) 3.8 A counterclockwise

9. An LC circuit has a capacitance of 20 $\mu$F and an inductance of 10 mH. At time $t = 0$ the charge on the capacitor is 27 $\mu$C and the current is 80 mA. The maximum possible current in mA is:

(1) 100 (2) 120 (3) 140 (4) 160 (5) 180

10. A horizontal power line carries a current of 4.0 kA from east to west. Earth's magnetic field of magnitude 50 $\mu$T is directed toward the north and is inclined downward from horizontal at an angle of 30°. What is the magnitude of the magnetic force, per unit length, on the power line?

(1) 0.20 N/m (2) 0.10 N/m (3) 0 (4) 0.50 N/m (5) 0.40 N/m

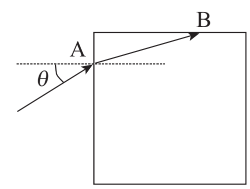
11. A plane electromagnetic wave in vacuum has $\vec{E} = (2\ \text{V/m})\cos[\omega(t - x/c)]\hat{k}$. What is the associated magnetic field?

(1) $\vec{B} \approx -7\ \text{nT}\cos[\omega(t - x/c)]\hat{\jmath}$
(2) $\vec{B} \approx +7\ \text{nT}\cos[\omega(t - x/c)]\hat{\jmath}$
(3) $\vec{B} \approx -2\ \text{T}\cos[\omega(t - x/c)]\hat{\jmath}$
(4) $\vec{B} \approx +2\ \text{T}\cos[\omega(t - x/c)]\hat{\jmath}$
(5) Insufficient information

12. A laser beam with intensity $4.3 \times 10^6$ W/m$^2$ and wavelength 632.8 nm is aimed vertically upward. What is the maximum radius in nm of a spherical particle (density 4100 kg/m$^3$) that can be supported by the laser beam against gravity ($g = 9.8$ m/s$^2$)? Assume that the particle is totally absorbing.

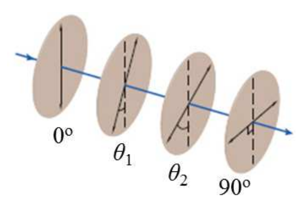
(1) 270 (2) 490 (3) 1070 (4) 120 (5) 190

13. A ray of light enters from air into a glass cube at point A at an incident angle of $\theta = 30°$ and then undergoes total internal reflection at point B. What minimum value of the index of refraction $n$ of glass can be inferred from this information?



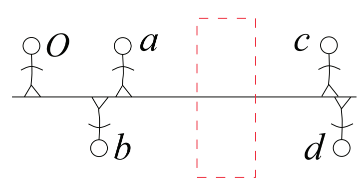

(1) 1.12 (2) 1.39 (3) 1.41 (4) 1.22 (5) 1.32

14. Unpolarized light is incident on four polarizing sheets with their transmission axes oriented as shown in the figure, where $\theta_1 = 40°$ and $\theta_2 = 65°$. All angles are between the transmission axis and the vertical, and are not necessarily drawn to scale. What percentage of the initial light intensity is transmitted through this set of polarizers?




(1) 19.8% (2) 15.8% (3) 19.4% (4) 17.5% (5) 21.1%

15. As shown, stick figure $O$ stands in front of a thin lens that is mounted within the boxed region. The solid horizontal line indicates the central axis of the lens. The four stick figures $a$ to $d$ suggest *general locations and orientations* for the images that might be produced by the lens. (Neither their height nor their distance from the lens is drawn to scale.) Which of the stick figures could not possibly represent images?




(1) $b$ and $c$ (2) $b$ and $d$ (3) $a$ and $d$ (4) $a$ and $c$ (5) only $d$

16. Two identical converging lenses with a focal distance of $f = 8$ cm are separated by a distance of 16 cm. An object of 1-cm height is placed 2 cm in front of the two-lens package. What is the absolute value of the image height formed by the two lenses?

(1) 1 cm (2) 0.7 cm (3) 1.4 cm (4) 0.5 cm (5) 2 cm

17. A person cannot see clearly objects closer than 1.75 m from her/him. What is the refractive power (in diopters) of the correction lenses that will allow her/him to clearly see objects located 25.0 cm, but not closer, from her eyes? Ignore the distance between her/his eyes and the lenses. Diopter is 1/f, where f is a lens's focal distance in meters.

(1) 3.43 (2) 2.83 (3) 2.78 (4) 3.04 (5) 3.27

18. Two slits of width 0.5 $\mu$m and separation 0.12 mm are illuminated by a coherent beam of light of wavelength 600 nm. What is the linear separation of the bright interference fringes observed on a screen that is a distance 5 m away?

(1) 25 mm (2) 0.04 mm (3) 2.0 mm (4) 10 mm (5) 5.0 mm

19. A lens is made from a material with an index of refraction of 1.5. It is coated with a thin film material to reduce the reflection of light of $\lambda = 532\,\text{nm}$. The index of refraction of the film material is 1.3. Which layer thickness in nm would minimize the reflectivity at that wavelength.

(1) 102 (2) 532 (3) 266 (4) 205 (5) 87

20. The Thirty Meter Telescope will be commissioned in year 2022. Its reflective mirror will be 30 m in diameter, as the telescope name suggests. In December 2022, Mars will be at its closest approach to Earth of about 80,000,000 km (such close approaches happen approximately once in three years). Which of the following object sizes will the telescope be able to resolve on Mars? The wavelength of visible light ranges from 400 to 700 nm.

(1) 2 km (2) 200 m (3) 20 m (4) 2 m (5) 20 cm